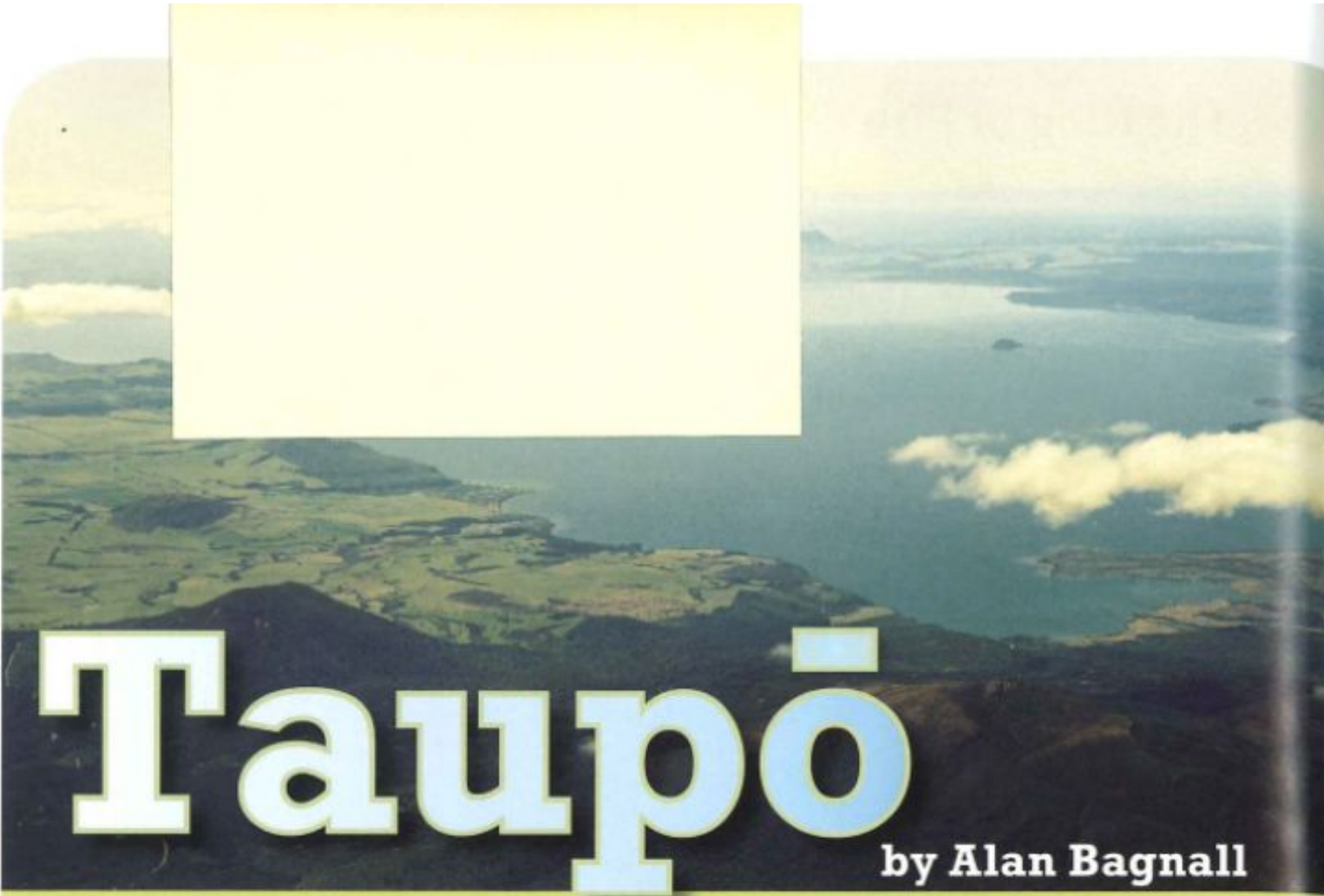


An aerial photograph showing a vast, turquoise lake (Lake Taupō) surrounded by dark, forested hills and green fields. The sky is blue with some white clouds. A large, solid yellow rectangle is positioned in the upper left corner of the image.

Taupō

by Alan Bagnall

In the centre of the North Island, there is a huge caldera volcano – Lake Taupō. Lake Taupō was formed by two massive eruptions.

The Oruanui Eruption

The first Taupō eruption took place 26 500 years ago. This was one of the biggest eruptions in the world during the last million years. The Oruanui eruption (as it is known) probably caused darkened skies all over the world for several years as ash blotted out the light from the Sun. Plants would have stopped growing, and people and animals would have had little food. People who lived on Earth then would not have known what was causing these changes.



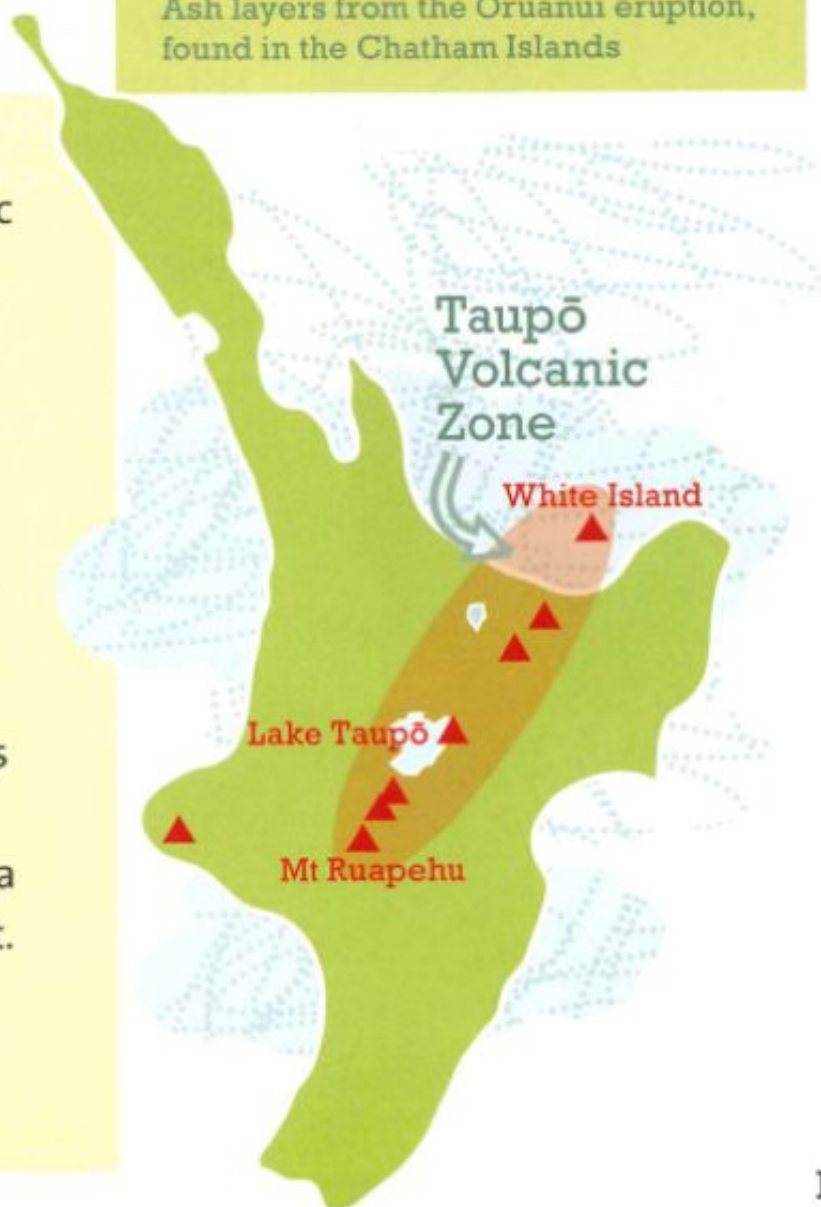
How do we know? Ash layers from volcanoes and eruptions have different textures (how they feel), colours (what they look like), and mixtures of chemicals (what they are made of). This is how geologists can tell which volcano ash has come from. Ash from the Oruanui eruption has been found in Chile! On the Chatham Islands, there is a layer of Oruanui ash that is 6 centimetres thick. Measuring the depth of an ash layer and how far it has spread helps geologists to work out the size and force of an eruption.



Ash layers from the Oruanui eruption, found in the Chatham Islands

Pieces of burnt wood and charcoal are buried within volcanic ash layers. Wood and charcoal decay (or rot) and change over time. Scientists work out how old this material is by measuring how fast it decays. They do this by a process called carbon dating. Carbon dating gives us a date for the Oruanui eruption.

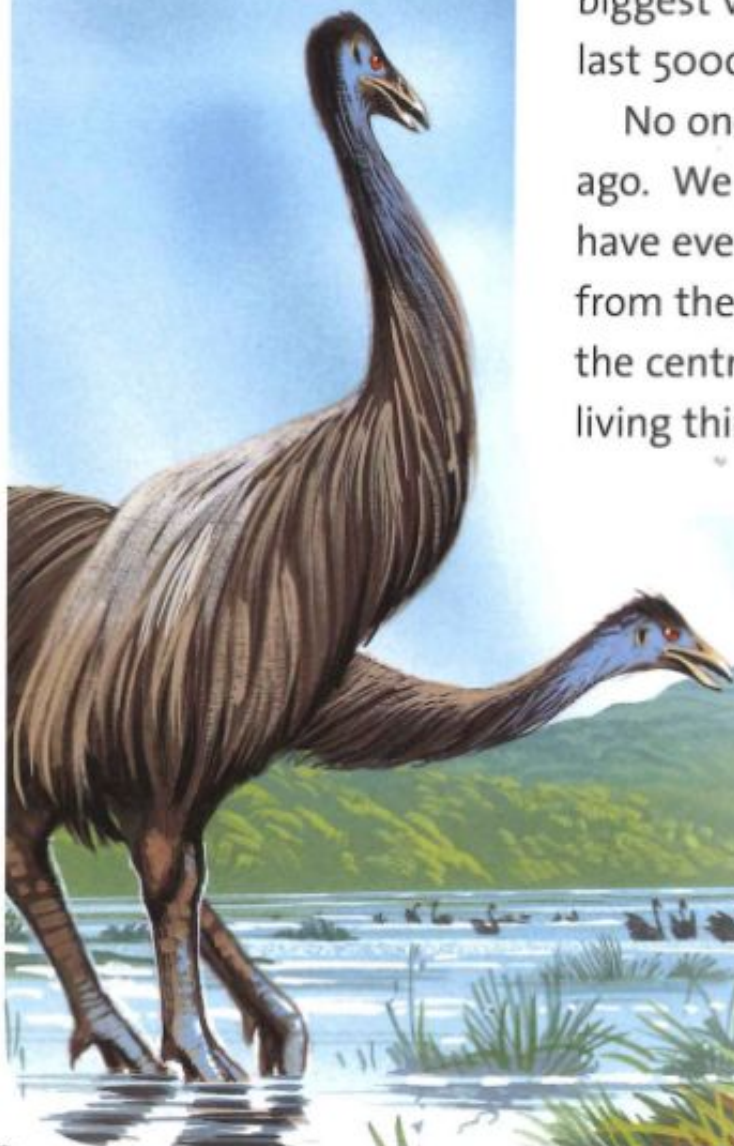
So much molten material was thrown out of the Taupō volcano during the Oruanui eruption that a great hole was left in Earth's crust. This hole slowly filled with water and became Lake Taupō.



The Most Recent Taupō Eruption

The most recent Taupō eruption was about 1800 years ago. At this time, the Chinese and Roman empires were flourishing. Chinese and Roman writings from that time record a bleak year when crops didn't ripen and a red dust darkened the air and blocked out the Sun. Of course, in China and in Europe, people had no idea that a volcano thousands of kilometres away in the South Pacific was causing these problems. The Taupō eruption is one of the biggest volcanic eruptions in the world in the last 5000 years.

No one was living in Aotearoa 1800 years ago. We know this because no human artefacts have ever been discovered in the ash layers from the eruption. The eruption devastated the central North Island. The forests and all living things were burnt and buried.

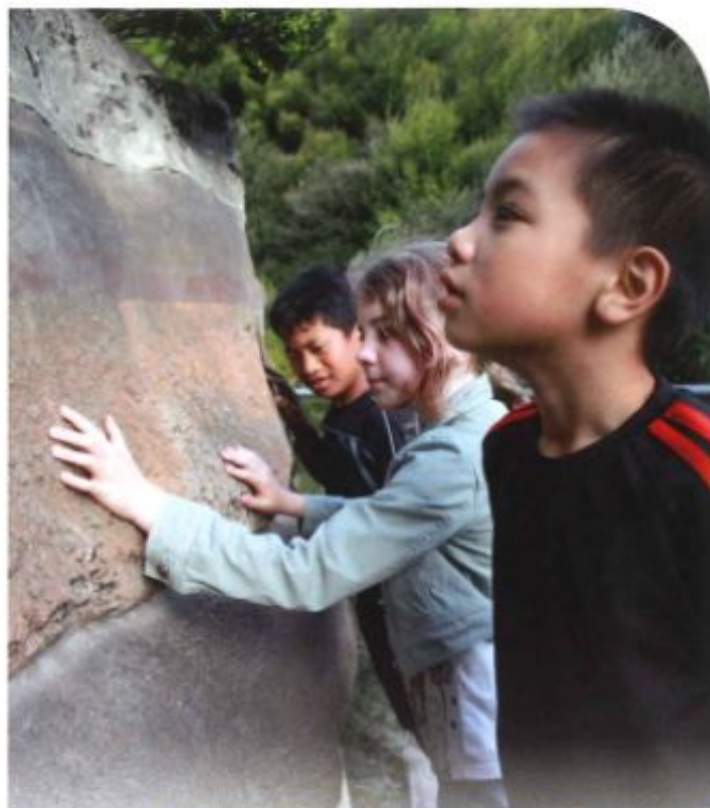


**Ruapehu
1995–1996**



How do we know the size of the Taupō eruption? Thick ash from the eruption can be seen in road cuttings around the central North Island. Scientists have carefully measured the depth of the layers to work out the force of the eruption and how long it lasted. They have also discovered that the Taupō volcano has had over twenty smaller eruptions at different times.

Comparing the sizes of eruptions



Students looking at ash layers

**Krakatoa,
Indonesia
1883**

**Taupō
AD 186**

**Vesuvius, Italy
AD 79**



VOLCANIC ROCKS

Rocks that have come from volcanic eruptions are named according to their mix of chemicals and their texture. The rocks that come from the Taupō eruptions are:

Tephra

Tephra is volcanic ash and debris that has fallen from the air and formed layers. Layers of tephra criss-cross the Central North Island.

Ignimbrite

Ignimbrite is formed from a hot ash flow that has hardened into rock. Ignimbrite cliffs can be found around Lake Taupō.

Obsidian

Obsidian is a natural volcanic glass found in the Taupō Volcanic Zone. When it's chipped and flaked, obsidian has a harder edge than steel. Māori valued obsidian very highly for making cutting tools. They traded it among tribes throughout Aotearoa.

Pumice

Pumice is a kind of magma with lots of tiny bubbles of gas trapped in it. When flung out of the crater, it cooled rapidly, trapping the gas bubbles and becoming pumice – that curious, floating stone found in rivers and lakes.

Rhyolite

Lava that contains a lot of silica is often sticky and thick. When this type of lava cools, it forms a rock called rhyolite.



Taupō's Human History

Ngāti Tūwharetoa are the guardians of Lake Taupō. The depths of the lake and the tops of the Tongariro volcanoes are tapu to them. In early times, the people of Tūwharetoa wore parekawakawa (a headdress of greenery) to avoid looking up at the mountains of Tongariro, Ruapehu, and Ngāuruhoe because their awe and respect for these sacred mountains was so great.

In 1846, Te Rapa village on the shores of Lake Taupō was buried when heavy rain caused a landslide of volcanic pumice and ash. The great Tūwharetoa chief, Te Heuheu II, and 60 villagers died.

Today Lake Taupō is a tourist centre. People enjoy boating and fishing on the lake and climbing and skiing on the nearby volcanoes. The mighty forces that made this wonderful landscape also make it a great place for scientists to study the Earth. A group of scientists called vulcanologists are studying the Taupō Volcanic Zone. The things they discover will help increase our knowledge of how volcanoes are formed and change over time.

